

S1 2024 Earnings Call

Company Participants

- Christophe Fouquet, President and Chief Executive Officer
- Roger Dassen, Executive Vice President and Chief Financial Officer
- Skip Miller, Vice President of Investor Relations

Other Participants

- Adithya Metuku, Analyst, HSBC
- Alexander Duval, Analyst, Goldman Sachs
- Andrew Gardiner, Analyst, Citigroup
- C.J. Muse, Analyst, Cantor Fitzgerald
- Chris Caso, Analyst, Wolfe Research
- Didier Scemama, Analyst, Bank of America
- Francois Bouvignies, Analyst, UBS
- Krish Sankar, Analyst, TD Cowen
- Mehdi Hosseini, Analyst, Susquehanna
- Sandeep Deshpande, Analyst, JP Morgan
- Stephane Houri, Analyst, ODDO BHF
- Tammy Qiu, Analyst, Berenberg

Presentation

Operator

Good day, and thank you for standing by. Welcome to the ASML 2024 Second Quarter Financial Results Conference Call on July 17, 2024. At this time, all participants are in a listen-only mode. After the speakers' introduction, there will be a question-and-answer session. (Operator Instructions) Please be advised that today's conference is being recorded.

I would now like to hand the conference call over to Mr. Skip Miller. Please go ahead.

Skip Miller {[BIO 20244900 <GO>](#)}

Thank you, operator. Welcome, everyone. This is Skip Miller, Vice President of Investor Relations at ASML. Joining me today on the call are ASML's CEO, Christophe Fouquet; and our CFO, Roger Dassen.

The subject of today's call is ASML's 2024 second quarter results. The length of this call will be 60 minutes and questions will be taken in the order that they are received. This call will also be broadcast live on -- over the Internet at asml.com. A transcript of management's opening remarks and a replay of the call will be available on our website shortly following the conclusion of this call.

Before we begin, I'd like to caution listeners that comments made by management during this conference call will include forward-looking statements within the meaning of the federal securities laws. These forward-looking statements involve material risks and uncertainties. For discussion of risk factors, I encourage you to review the safe harbor statement contained in today's press release and presentation found on our

website at asml.com, and in ASML's annual report on Form 20-F and other documents as filed with the Securities and Exchange Commission.

With that, I'd like to turn the call over to Christophe Fouquet for a brief introduction.

Christophe Fouquet {[BIO 20524437](#) <GO>}

Thank you, Skip. Welcome, everyone, and thank you for joining us for our second quarter 2024 results conference call. Before we begin the Q&A session, Roger and I would like to provide an overview and some commentary on the second quarter 2024, as well as provide our view of the coming quarters. Roger will start with a review of our second quarter 2024 financial performance with added comments on our short-term outlook. I will complete the introduction with some additional comments on the current business environment and on our (Technical Difficulty) outlook.

Roger?

Roger Dassen {[BIO 15064806](#) <GO>}

Thank you, Christophe, and welcome, everyone. I will first review the second quarter 2024 financial accomplishments and then provide guidance on the third quarter of 2024.

Let me start with our second quarter accomplishments. Total net sales come in at EUR6.2 billion, which is just above guidance. Net system sales of EUR4.8 billion, which is made up of EUR1.5 billion of EUV sales and EUR3.3 billion of non-EUV sales. Net system sales was driven by logic at 54%, with the remaining 46% coming from memory. Installed base management sales for the quarter came in at EUR1.48 billion, slightly higher than guided.

Gross margin for the quarter came in at 51.5%, which was above our guidance, primarily driven by more immersion systems than planned.

On operating expenses, R&D expenses came in slightly above guidance at EUR1.1 billion, due to higher R&D expenses regarding metrology and inspection, while SG&A expenses came in slightly below our guidance at EUR277 million as a result of lower IT spend. Net income in Q2 was EUR1.6 billion, representing 25.3% of total net sales and resulting in an EPS of EUR4.01.

Turning to the balance sheet. We ended the second quarter with cash, cash equivalents and short-term investments at a level of EUR5.0 billion. We ended Q2 with free cash flow of EUR386 million, which is an improvement relative to last quarter. However, as mentioned last quarter, there continues to be pressure on free cash flow as we provide some support for our customers and operate at higher inventory levels, the latter being a result of the increased material intake including High NA as part of our planned capacity ramp in preparation for strong demand next year. We expect a gradual return to normal cash conversion levels as the industry continues to recover.

Moving to the order book. Q2 net system bookings came in at EUR5.6 billion, which is made up of EUR2.5 billion of EUV bookings and EUR3.1 billion of non-EUV bookings. Net system bookings in the quarter were mainly driven by logic at 73% and memory the remaining 27% of the bookings. At the end of Q2 2024, we finished with a backlog of around EUR39 billion.

With that, I would like to turn to our expectations for the third quarter of 2024. We expect Q3 total net sales to be between EUR6.7 billion and EUR7.3 billion. We expect our Q3 installed base management sales to be around EUR1.4 billion. Gross margin for Q3 is expected to be between 50% and 51%. We still expect a slightly lower gross margin for the full year 2024 compared to 2023. Although margins for the second half of the year are expected to be positively impacted by higher volume and more favorable EUV Low NA mix, we expect this to be offset by increased High NA costs relative to the first half of the year.

The expected R&D expenses for Q3 are around EUR1.1 billion and SG&A is expected to be around EUR295 million. Our estimated 2024 annualized effective tax rate is expected to be between 16% and 17%.

In Q2, ASML paid a final dividend of EUR1.75 per ordinary share. Together with the interim dividends paid in 2023 and 2024, this resulted in a total dividend for 2023 of EUR6.10 per ordinary share. The first quarterly interim dividend over 2024 will be EUR1.52 per ordinary share and will be made payable on August 7, 2024. In Q2 2024, we purchased 106,000 shares for a total amount of EUR96 million.

With that, I would like to turn the call back over to Christophe.

Christophe Fouquet {[BIO 20524437 <GO>](#)}

Thank you, Roger. As Roger highlighted, a good quarter with sales and profitability above guidance. Although macro uncertainty remains, overall semiconductor industry levels continue to improve, trending towards more healthy levels. We also see continued improvement in lithography tool utilization level at both logic and memory customers, all in line with the industry's continued recovery from the downturn.

Our outlook for the full year is unchanged with revenue expected to be similar to last year. We currently see strong developments in AI driving most of the industry recovery and growth ahead of other end market segments. We still expect a stronger second half relative to the first half of the year.

Logic customers continue to digest the significant capacity additions made over the past year. With this digestion, we see lower revenue from logic this year relative to last year. In memory, demand is primarily driven by DRAM technology node transitions in support of advanced memories such as DDR5 and HBM. In support of this transition, we expect growth in revenue from memory this year relative to 2023.

For our installed base business, we continue to expect a similar level of revenue compared to last year. As the recovery becomes more clear this year, customers may look to upgrade their system in preparation for 2025.

Turning to our businesses. We expect EUV revenue growth in 2024. We plan to recognize revenue on a similar number of EUV systems as 2023 with expected revenue from one to two High NA systems. On our Low NA EUV systems, (Technical Difficulty) additional NXE:3800E system this quarter as we continue to ramp production. As customers transition to advanced nodes in both logic and memory, we expect the majority of shipments in the second half of the year to be NXE:3800E system.

Regarding High NA, we shipped the second system this quarter, and it is currently under installation. The first system shipped to a customer is running qualification wafer.

Overall, the interest is high with EUV customers now using our High NA system in the joint ASML-imec demo lab in Veldhoven for their initial development.

We have already exposed wafer for multiple logic and memory customers and have now achieved the imaging resolution specification of 8 nanometer. High NA will enable an almost three time increase in transistor density relative to the Low NA system. So all in all, good momentum on High NA and progressing well against customer expectation.

We expect our non-EUV business to be down in 2024, primarily driven by lower emerging system sales relative to 2023. As we have stated over the last few quarters, we view 2024 as a transition year and continue to make investments this year, both in capacity ramp and in technology, to be ready for future demand. While the slope of the industry recovery is still uncertain, based on discussion with our customer and supported by our strong backlog, we expect 2025 to be a strong year, driven by a number of factors as mentioned last quarter.

First, the secular growth driver in semiconductor end markets which we have previously discussed, such as energy transition, electrification and AI are still very much intact. The expanding application space, along with increasing lithography on future technology nodes, drives demand for both advanced and mature nodes. Second, the industry expects to be in a cyclical upturn in 2025.

And last, as mentioned earlier, we need to prepare for a number of new fabs that are being built across the globe, in some instances clearly supported by several government incentive plans. These fabs are spread geographically and are strategic for our customers. They are all scheduled to take our tools.

It is essential that we keep our focus on the future and build capacity in preparation for future long-term growth, as we discussed in the market scenario for 2025 and 2030 during our Investor Day in November 2022. We plan to update our view during our Investor Day this year on November 14, 2024.

In summary, although there is near-term uncertainty, we remain confident in our long-term growth opportunity.

With that, we will be happy to take your questions.

Skip Miller {[BIO 20244900 <GO>](#)}

Thank you, Roger and Christophe. The operator will instruct you momentarily on the protocol for the Q&A session. Beforehand, I would like to ask you that you kindly limit yourself to one question with one short follow-up, if necessary. This will allow us to get to as many callers as possible.

Now, operator, could we have your final instructions and the first question, please?

Questions And Answers

Operator

Thank you. (Operator Instructions) We will now go to the first question. And your first question comes from the line of Krish Sankar from TD Cowen. Please go ahead.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Yeah. Hi. Thanks for taking my question. Christophe or Roger, the first question is on the bookings composition and the outlook. In the booking for EUV, in the EUR2.5 billion, were there any 2-nanometer orders from foundry customers that you said you should expect in Q2 or Q3? Any High NA in the mix? And based on the bookings momentum and backlog, how do you think about calendar '25 revenues? Do you think it could be towards the upper end of the EUR30 billion to EUR40 billion range? Then I had a quick follow-up.

A - Roger Dassen {[BIO 15064806](#) <GO>}

So on the bookings, as you would have seen in the bookings mix, 73% of the bookings is related to -- was related to logic. So I think that gives you a pretty strong indication who the buyers are and who is in the mix. So I think it is reasonable to assume that the foundry business at 2 nanometer is in the book. So that's one.

High NA is not in there. So no High NA bookings in this quarter. So everything you see there on EUV is related to Low NA.

As it relates to 2025, Krish, we're very consistent, I think, in our messaging, and that's what we say. We confirm what we said in November of 2022, which is we expect 2025 to be between EUR30 billion and EUR40 billion. We've also said it's not the low point of that -- of that guidance. Last quarter, we've given you a bit of an -- a bit of an understanding of what it takes to be fully booked at the beginning of 2025 for the midpoint of the range. I think you can conclude with the bookings number that, that we've reported that we're nicely -- that we're nicely on track.

But with that, we still confirm that our expectation is between the EUR30 billion and EUR40 billion and are not going to be more specific than what we've been so far.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Got it. Roger, it's very helpful. (Multiple Speakers)

A - Roger Dassen {[BIO 15064806](#) <GO>}

Krish, one more thing, and Christophe was very clear on that in his comments. I think in the Capital Markets Day that we're going to have in November of 2024, that's going to be the point in time where we are going to be more specific. So that's where we will be more specific as to what we expect for 2025. That's when we're going to narrow the guidance, but we're not going to do that before that.

Q - Krish Sankar {[BIO 16151788](#) <GO>}

Thanks, Roger. Thanks a lot for that amazing color. And then a quick follow-up for Christophe. Obviously, the article today about the US administration looking to impose some severe trade restrictions using the foreign direct product rule on China shipments. I know, obviously, you folks are good citizens, follow the rules. Kind of curious what do you think about the potential implication for ASML from here? How much more severe could it be, given the fact that you do have a pretty high exposure to China? If you can give some color or some bookends around it, that'd be very helpful. Thank you.

A - Christophe Fouquet {[BIO 20524437](#) <GO>}

Well, I think first, we don't comment on rumors, and there's a lot of rumors on that topic. So I think on that topic, we comment even less, I will say. So that's the first thing.

I think we had discussion on China in the past together. I think the point Roger and I have made in the last few quarters, which is still valid, is before you look at China, I think you have to look at the opportunity we see on the mature semiconductor market. That opportunity was defined as being significant in our Capital Market Day in November 2022. We still believe it's significant.

In 2023, 2024, China has been investing a large part of this market. And this is why I think our revenue on China has been high. Moving forward, we still believe that this market is needed and therefore that the capacity (Technical Difficulty) coming out of somewhere. So that's a bit the position we take on that discussion. I think we always look at the end market, at what's the need of the end market. I think we are a bit less sensitive on where that capacity may be produced.

Q - Krish Sankar {BIO 16151788 <GO>}

Thanks, Christophe. Thank you.

Operator

Thank you. We will now take the next question, and the question comes from the line of Alexander Duval from Goldman Sachs. Please go ahead.

Q - Alexander Duval {BIO 16682293 <GO>}

Yeah. Thanks for the question. You confirmed today that it's fair to assume that a logic customer ordered on 2 nanometer this quarter. But clearly for such a large node, one would expect to see more order flow. So I wonder to what extent, it's fair to assume one should expect more EUV orders from them in the coming quarters.

And as a quick follow-up, you mentioned in your prepared remarks that AI is driving most of the industry's recovering growth. Are you able to decompose? What specifically is driving this for ASML? To what extent, do you see litho needs being driven by high bandwidth memory, data center processing, or edge AI? Is there any area that's surprising to the upside? Many thanks.

A - Roger Dassen {BIO 15064806 <GO>}

It's clear, I think, that if you look at the orders that came in and there is a healthy start, I would say, for the M2[ph] foundry business, I think it's a healthy start. It's also clear that obviously not the full capacity that you would need for this node is in there, but that is pretty logical, right? I mean, obviously, this customer is going to ramp, and there you will see a gradual buildup, and that will -- you will also see -- at some stage, you will see additional orders coming in for that ramp, exactly what that buildup will be you might want to raise that question by tomorrow.

And maybe the second part, I think, is for you, Christophe.

A - Christophe Fouquet {BIO 20524437 <GO>}

Yeah, I think so -- I think I said it in my comment initially. I think we've been also consistent there. So I think AI is driving, I would say, right now, the biggest part of the recovery. This is true for logic. This is true for memory. Roger just commented on logic. I think you know that for high bandwidth memory, those products drive more demand, more wafer demand, because we are looking basically at a higher density of DRAM on

those products, and we look at something that, of course, will take course over several months. So we started to see the positive effect of that for 2025.

We expect that to continue into 2026, both for memory and for logic. And at some point of time, also mentioned that maybe the other segments are a bit behind in terms of recovery. So a lot of the capacity today, either logic or DRAM capacity, will be (Technical Difficulty) those AI product. As the other segments recover, we also expect potentially some capacity to be needed there. And that's also why we believe overall, that 2025 will be a strong year. And I think what everyone is trying to understand, what you try to understand is exactly the pace of it that I think we don't know exactly, but the fundamentals, I would say, on AI and I think the recovery in general are strong from our point of view.

Q - Alexander Duval {[BIO 16682293](#) <GO>}

Many thanks.

Operator

Thank you. Your next question comes from the line of Tammy Qiu from Berenberg. Please go ahead.

Q - Tammy Qiu {[BIO 17604871](#) <GO>}

Hi. Thank you for taking my question. So, first one on the leading-edge 2-nanometer ramp-up pace. So you mentioned that they are starting to ramp-up their leading-edge 2-nanometer capacity already, or preparing for that. Do you notice anything in this quarter suggesting that they are pulling in the ramp-up phase or everything is as planned?

And the second question I have is regarding the DRAM adoption of more EUV layers as well as High NA EUV, how do you see the layer count change over the next few generations? Are we going to see similar level of growth of layers as we have seen over the past two years, or it will gradually slow down? If there is a number you can share, it will be great. Thank you.

A - Roger Dassen {[BIO 15064806](#) <GO>}

I'll take the first part, Christophe, and you can take the second. But as it comes to, do you notice any pull in there, I think the foundry customer has indicated that they're looking at M2[ph] in the second half of 2025. I think you're typically for this customer looking at a lead time of around a year. As we've said before, I think putting in orders at this stage I think is consistent with that to the extent that there is a difference in timing or ramp. Again, I think that's a question that you might want to raise on a conference call tomorrow. But I think what we see here is pretty consistent with what we've also heard publicly -- heard stated publicly about starting this in the second half of 2025.

Christophe, on DRAM?

A - Christophe Fouquet {[BIO 20524437](#) <GO>}

Yeah. On DRAM, so I think they also I think be very consistent with the information we have shared with you previously. So we see on the -- an increase of EUV use on every node. I think this is a trend that continue at least in the foreseeable future. Of course, it's always more difficult to make forecasts on nodes or technology that are still being

defined by customer, but that logic is still in place. I think you have seen also in DRAM that this point of time, all customers are using EUV in production. I think the last customer was very public about that recently. So you see a gradual basically adoption with different speed, maybe per customer, but I would say very much in line with I think what we explained already now for a few years.

On High NA, we also see opportunity for DRAM at the horizon of '25, '26. As I mentioned in my commentaries, we see customers very eager to test the High NA tool in our lab, exposing their own wafer to validate basically that High NA can provide a nice opportunity both in performance, but also in cost savings. So they also same as before. We believe that '25, '26 is about the time where we could see some insertion of High NA for DRAM.

Q - Tammy Qiu {[BIO 17604871 <GO>](#)}

Okay, thank you.

Operator

Thank you. We will now take the next question. And your next question comes from the line of Francois Bouvignies from UBS. Please go ahead.

Q - Francois Bouvignies

Thank you very much. My first question would be on China, I'm afraid, I mean, and it's very simple. Could you make in the future deep EUV tools without US IP at all? And are all your deep EUV tools using US IP? That's my first question.

A - Roger Dassen {[BIO 15064806 <GO>](#)}

That's a very hypothetical question. At this stage, there is -- as you know, we have a lot of operations also in the United States, so there is a significant contingent of US technology in those tools. So speculating about could we do it without US technology, I think, is speculation that we shouldn't -- that we shouldn't and will not go into, I think, or the way we produce and the way we run our business should not be influenced by this kind of question.

Q - Francois Bouvignies

Okay.

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

Yeah, I think -- maybe to add to that, I think we've been also very vocal to say that we believe that preserving the ecosystem as much as possible is a good thing for this industry. I think we still believe that. And therefore, I think that's still also the discussion we're trying to have with all the stakeholders.

Q - Francois Bouvignies

Makes sense. Thank you very much. And maybe the second question I had is a bit of a follow-up. So if we take like this EUV orders of 2.5 billion and your 39 billion of backlog that you have, I come with like, you still have 30 tools -- EUV tools to book for next year to reach the midpoint of the guidance. So, Roger, you said nicely that your order backlog

is -- orders is going well on track above this 4 billion, but it's mainly driven by deep EUV. And on EUV, it seems that you still need 30 to book for the rest of the year, which would imply around 7 billion of EUV orders coming in H2 versus 3 billion in H1. Is my math correct in a way to think that you still have a significant part of your EUV to be booked in the second half of the year, assuming, of course, you have one year lead times?

A - Roger Dassen {BIO 15064806 <GO>}

There's still -- I think the math that we gave before, I think, still holds. So as I mentioned, I think we're well on track to achieve the objective -- the objective there. And we have given indications or we have given a bandwidth for 2025, Francois, for the full revenue. I think you're now jumping into one specific element of the revenue, but I think our math holds for the full revenue, and I think that's what holds. And I think going deeper into that and going there I don't think adds anything to the overall message that I believe we are nicely underway to achieve the objectives and the bandwidth that we've discussed before. So I wouldn't go -- I wouldn't want to go at this stage any more specific into any of the components. As I mentioned before, we have a Capital Markets Day coming up. There we will be a lot more specific on how we look at 2025. And until that point in time, I think we're just going to stick to our core messaging that I've given before.

Q - Francois Bouvignies

Understood. I tried my chins. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

You did.

Operator

Thank you. We will now take the next question. And your next question comes from the line of Stephane Houri from ODDO BHF. Please go ahead.

Q - Stephane Houri {BIO 3899118 <GO>}

Yes, hello. Thank you for taking my question. In your prepared remark this morning, you seem to talk pretty much about increasing capacity or at least preparing for the ramp. So does it mean that -- and at the same time, you're talking about -- very highly about the reception of EUV High NA. So maybe if you could clarify what you mean by preparing the ramp and if it does mean that you want to increase your capacity, do more investments and maybe open a new plant. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

So in general, when we talk about increase in capacity, we really talk about increasing capacity across the board. And this really relates to the capacity expansion that we've talked about in the past, which is that we're driving towards the capacity of 600 deep EUV tools and 90 EUV tools. But then added to that, we are indeed also expanding our capacity when it comes to High NA. And that's I think -- it's that combination. So when we talk about adding capacity and ramping capacity, it's really across the board, it's the 90, 600[ph]. And as we mentioned before, medium term working towards 20 High NA tools.

Q - Stephane Houri {BIO 3899118 <GO>}

Okay. And can you maybe clarify where we're standing in term of ASP for High NA tools? Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

On the High NA tools, what we said before is that we're looking at an ASP north of EUR350 million. That's the ASP that we have for the 5200[ph] tool, which is the high volume manufacturing tool that customers are ordering in this time frame.

Q - Stephane Houri {BIO 3899118 <GO>}

Okay. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

You're welcome.

Operator

Thank you. Your next question comes from the line of Andrew Gardiner from Citi. Please go ahead.

Q - Andrew Gardiner {BIO 7137663 <GO>}

Hi. Good afternoon, gentlemen. Thanks for taking the question. If you don't mind, I'll try another one on China. I'll try and be specific and we'll see whether you can comment on it. The article this morning seems to be, again, quite focused on your service activity in China and in particular, your servicing of tools that you were able to ship last year, high-end immersion tools, NXT:2050i[ph] models. But as of the start of this year, you're no longer able to ship. Can you tell us whether you're still indeed servicing those tools at some of those clients? Yeah. Can you just clarify that?

A - Roger Dassen {BIO 15064806 <GO>}

Well, we are following all the applicable laws and regulations when it comes to export controls. And of course, that is a landscape with US rules and with Dutch rules. So for some fabs, there are more limitations to others. But in general, it is still the case that we can operate in the fabs of our customers. So we still have -- as we call it, we still have eyes on the tools, so we're still in the fabs of our customers. But there are limitations that for some fabs are more stringent, and this is then particularly related to parts being sent either from the United States or from Europe or when it comes to the use of technology, so manuals, et cetera. So there are limitations that for some fabs are more stringent than for other fabs. But in general, we're still in a position that our people can continue to serve in the fabs of our customers in the (Technical Difficulty).

A - Christophe Fouquet {BIO 20524437 <GO>}

And I think we'll be aware we have been now in this position for quite a few months. I think the landscape has been set. So I think Roger said it. We fully understand the new regulation. We, I would say, adjusted our service, our action according to that. But for us, this is not new news at this point of time. We have been doing that for many months already.

Q - Andrew Gardiner {BIO 7137663 <GO>}

Thank you. Also, if I could have a quick follow-up. Christophe, you mentioned the new fabs that are going to be built, some of which supported by CHIPS Act money or equivalent funds in other jurisdictions. With all of those products scheduled to take tools, how much of an impact do you think is that in 2025 and how much is actually beyond that, given the lead time it takes to build a lot of concrete and steel going in at the moment, but there's certainly not many ready to take actual clean room equipment just yet? Can you help us with how you think that's going to phase over '25 and then '26 and beyond?

A - Christophe Fouquet {[BIO 20524437](#) <GO>}

Yeah. So I think to make the answer simple, I think most of it will come, in fact, after 2025. So I think the volume from those fab is still limited next year. I think you're mostly aware of the big announcement either in the US, in Europe, in Japan. So I think this will come after '25, mostly starting 2026. This was also with my comment about preparing also the capacity for that. So as Roger explained, when we prepare the capacity, taking into account the lead time we have for some of our factories, some of our suppliers, of course, the capacity is there to meet the market demand, in fact, beyond 2025. So this is what we wanted to be ready for.

So this is also why we are still very comfortable with the long-term view of the market, because those things are going to come. And we could expect even some more government funding moving forward, because as you also see it in the news, I think there is more and more wish from everyone to invest in this business. So I think that's also something we don't expect to slow down over time. We expect that there will be more after the first wave of investment.

A - Roger Dassen {[BIO 15064806](#) <GO>}

So Andrew, that's absolutely right. And to add to that, of course, there are many new fabs that are being built, and not all of them are necessarily associated with the CHIPS Act money. So that's why also in the past, we said that when it comes to 2025, more than half of the shipments will go (Technical Difficulty) but that will include new fabs -- including new fabs, for instance, to be -- that are being constructed in Taiwan. So more than 50% of our shipments for 2025, we expect to go to new fabs. But Christophe is absolutely right. I think the effect really of the CHIPS Act money, I think you should probably see beyond that year.

Q - Andrew Gardiner {[BIO 7137663](#) <GO>}

Thank you very much.

Operator

Thank you. We will now take the next question. And the question comes from the line of Chris Caso from Wolfe Research. Please go ahead.

Q - Chris Caso {[BIO 4815032](#) <GO>}

Yes, thank you. Good morning. The question is regarding revenue for the back half of the year. And based on where you're guiding for the third quarter and maintaining the view on the full year, that does anticipate some acceleration as you go into the fourth quarter. Could you give some color on what's going on there with regard to the timing of the

orders? Is this just kind of typical some timing differences between some of the shipments in the third quarter and fourth quarter? And what gives you the confidence in the fourth quarter, please?

A - Roger Dassen {BIO 15064806 <GO>}

Chris, what you see is that we are indeed building up momentum. I think that's also what you might expect, given what we said we look at 2025 as a growth year. And I think what you see over the quarters in 2024 that it is clearly building up over the quarters. I think - and as we said before, we have been building capacity, so we're able to do it. But the demand is also building up this momentum over the quarters leading into the growth per year in 2025. I should also add that as we've mentioned before, there is also some deferred revenue that we've been building up. And that means systems that we have shipped that we have not yet recognized revenue for and the impact for that in the second half is expected to be around a billion[ph]. So that's a substantive number.

And that comes either from fast shipments that we have been doing. And as you know, we no longer intend to fast ship by the end of this year. So that gives you a bit of a tailwind there. But it's also a number of 3800 tools that we're shipping and that we take revenue for after shipments at site acceptance for the customer. And also, as you know, we're in the process of shipping High NA tools that have been shipped, but where we only take revenue at the moment of site acceptance test.

So there is a number of reasons why we will recognize revenue for stuff that has already been shipped. And I think with that tailwind and with the building up of momentum both in terms of capacity and in terms of demand, we're indeed looking at a very progressive buildup of revenue throughout the year, this year and leading to over 9 billion expectation indeed in Q4.

Q - Chris Caso {BIO 4815032 <GO>}

That's very helpful color. Thank you. As a follow-up, I just wanted to follow-up on some of the comments you spoke about for High NA DRAM. And speaking of (Technical Difficulty), you think '25, '26 insertion time frame for that. Obviously the lead time for those tools are very long. It sounds like the customers are in your labs right now. What's your anticipation regarding timing on development tools or production tools for that and some idea of perhaps magnitude of tools or layer count for that going forward?

A - Christophe Fouquet {BIO 20524437 <GO>}

Well, I think -- maybe I should start. I think as we said in the past, we have, I would say, a good backlog for High NA, which means that the customer I was referring to, all our EUV customers have ordered those systems, you see that we start to deliver those tools. And of course, this is timely aligned with some of the opportunity to insert.

The addition of the lab basically is good for our customer because this gives them a chance even before that tool hit the ground to get the first data. And I think you understand on the technology[ph] High NA, the very first data are critical to understand exactly what's the value of the technology. So this is why today the wafer that are being exposed are already providing very valuable information to our customer, information they can start to use to refine a bit their insertion scenario and the value that High NA could bring. That's why we've been quite excited about the fact that we are generating

this data today. And as I mentioned, I think what we see so far is the data are well meeting our customer expectations. So this is also possible[ph].

Q - Chris Caso {BIO 4815032 <GO>}

Thank you.

Operator

Thank you. Your next question comes from the line of C.J. Muse from Cantor Fitzgerald. Please go ahead.

Q - C.J. Muse {BIO 6507553 <GO>}

Yeah. Good morning. Good afternoon. Thank you for taking the question. I guess first question, would love to hear your thoughts around any changes in customer conversations over the last three months, whether foundry logic, and particularly as it relates to your more robust immersion shipment outlook for the second half of this year.

A - Christophe Fouquet {BIO 20524437 <GO>}

Well, the short answer is no. No big change. I think the discussion we have with our customer are very consistent. So a lot of ramp of AI, the ramp beyond that, we don't see any major change. I think, in fact, I would say our view of the market this quarter is very much the same as the one we had last quarter. I think the big difference, as you have seen, is that we have made progress with booking on some of the segments, answering maybe one of the big question we got from all of you last quarter. But the view of the market for Roger, for me, is not changing. I think the discussion are very (Technical Difficulty).

A - Roger Dassen {BIO 15064806 <GO>}

When it comes to immersion (Multiple Speakers). Go ahead, CJ.

Q - C.J. Muse {BIO 6507553 <GO>}

Sorry. Please go ahead.

A - Roger Dassen {BIO 15064806 <GO>}

When it comes to immersion, your other question, we continue immersion to continue to be strong, even though in the total mix, we do believe that the mix in the second half, there will be a lot more dry. So in the deep EUV mix, you will see the dry component becoming more significant than it was in the first half. But overall, we do believe that immersion will continue to be strong also in the second half of the year.

Q - C.J. Muse {BIO 6507553 <GO>}

Thank you. And as a quick follow-up, can you speak to the 3800? You talked about majority of shipments in the second half of this year. Think about the impact to gross margins and ASPs second half '24, and then kind of same question for all of 2025. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

So indeed, as Christophe said, it will be -- the majority of EUV tools in the second half will be 3800. And I should add, as you know, CJ, and we talked about this before, we have a ramp schedule where the initial 3800s are not yet up to full configuration. So gradually, we will get the 3800s up to the full specification of 220 wafer per hour. It also means that not all of the 3800s that will ship in the second half of this year will command the higher ASP, because some revenue is deferred up until the point where the tools have been upgraded to that higher specification.

So we'll not get the full tailwind, I would say, of the strong 3800 pricing in the second half, but it's definitely going to help. But you will see a stronger effect of that in 2025, where of course, the share of the 3800 is going to be even stronger. And there all the tools essentially will be in final specification. And we will also get some deferred revenue that will be recognized in (Technical Difficulty) from tools that have been shipped in 2024.

So if you look at the gross margin for the second half, of course, this is one element in there. Of course, we'll have a slight -- and that's a positive, I would say. And then we have a higher volume, obviously, in 2025 -- in the second half of 2024, as we just calculated, that's also a positive. A slightly negative is that in the deep EUV mix, as I mentioned before, the dry component is going to be more visible, right? So that's a bit of a drag on the gross margin.

And then we also -- in the second half, we're going to have the first revenue recognition of the High NA tools. And of course, on the one hand, that's great, because that means that we have the sign off of customers on the tool, as we expect, in the second half of this year. But you also know that the gross margin on that tool is not great. So that will also be a drag. So that's the reason why for the second half of this year, we expect a slightly lower gross margin than we expect -- than we were able to report in the first half.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Very helpful. Thank you.

A - Roger Dassen {[BIO 15064806 <GO>](#)}

Welcome.

Operator

Thank you. Your next question comes from the line of Didier Scemama from Bank of America. Please go ahead.

Q - Didier Scemama {[BIO 21301064 <GO>](#)}

Yes. Good afternoon. I have a couple of questions. First on High NA, maybe for Christophe, (Technical Difficulty) good news that good stuff that you told us about transistor density and the quality of the imaging on High NA. How are your conversations going with one of your customers that historically was a bit reluctant with that technology? I understand they came to visit you in Veldhoven, which is reasonably unusual. Just tell us a little bit about the feedback you got from that visit, whether you think the bid-ask spread between what they want and what you can deliver, how is that progressing. And I've got a follow-up.

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

Yeah. So first I'd like to say I think we never qualify any customer as being reluctant for High NA. I think this information came from somewhere else. I think the short answer is the data generation help. Again, we are exposing important wafer. We told you in the past, we are happy to get that amount of backlog on High NA because our customers were showing the trust in the technology without having any data at hand. So they were basically following our, I would say, analysis or our promise that this will help them moving forward. And they have been willing to invest for R&D quite heavily and by the way, even down pay some of those systems. So this has a huge commitment from the industry.

I think the data is helping tremendously because our customer can see data. I think that can be used also for potentially their customer to have a much better discussion around High NA. So I think the dynamic is good. And I would say the discussion is getting a lot more concrete. And I think we need to give a few more months to that discussion, so that everyone, based on data this time, have a bit of a better idea of what could be done moving forward with High NA and how quickly.

Q - Didier Scemama {[BIO 21301064 <GO>](#)}

Excellent. Thank you. My follow-up is a bit of just picking your brains, and I'm sure you've got like armies of PhDs learning simulations. But if we were to assume that there is some sort of AI, PC and smartphone refresh cycle pushing volumes for those combined markets to maybe a high single-digit volume growth in calendar year '25 or calendar year '26, do you think that there is enough capacity in DRAM, maybe NAND, and maybe to a lesser degree in logic to deliver that potential uplifting volumes?

A - Christophe Fouquet {[BIO 20524437 <GO>](#)}

Well, I think everyone is maybe trying to answer that question. I think it's very difficult to answer the question. I think what we have seen with AI is a major investment. from many companies in supercomputer in the ability basically to train model. What we still miss in AI, I think, is the emergence of end product. So I think today, there's not much revenue made on AI. There's just a lot of investment. What we see is that still that investment require a lot of capacity. I think you have seen some of our customers announcing also more capacity to be built before 2028.

And as I said before, if the rest of the application were to recover, what has been done so far, which is to reallocate some of the capacity to HBM, may become more difficult. So overall, I think we still expect as the rest of the market recover to see a need for more capacity. And that's, I think, also true for DRAM. Now to tell you at what speed, I think this is still a bit difficult. I think we said it a few times. I think the recovery is something we are looking at every day and there is also a lot of uncertainty. So this is why you still see us being, I would say, strong in our view of 2025, but also a bit cautious because there's still a lot of uncertainty around those opportunities.

Q - Didier Scemama {[BIO 21301064 <GO>](#)}

Brilliant. Thank you.

Operator

Thank you. We will now take the next question. And the question comes from the line of Mehdi Hosseini from Susquehanna. Please go ahead.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Yes. Thank you for letting me ask the question to[ph] follow-up. Christophe, is there anything you can share with me to give me confidence that 3800 is going to finally be on track? If I go back to your '22 Analyst Day, the transition was supposed to happen in '23, got pushed that to early '24 and now it's a big hockey stick coming with the 3800E shipment. Is there anything you can share with us in terms of throughput or any other thing that would help us with incremental confidence? And I have a follow-up.

A - Christophe Fouquet {BIO 20524437 <GO>}

Yeah. So I think there's two parts of that. I think the technology part where we are busy here in ASML to demonstrate basically the performance of the tool. We have seen very big progress there. We have measured 195 wafer per hour on the tool. We expect to see the final specification being measured also in the coming weeks. We don't see any hard showstopper, so we just see some work to be done. With the customer, I think some of the delay was also a choice we made to support the demand in the previous year. So if you remember, 2022 was a year where we saw a very strong demand from all customers on EUV. And we thought this would be wise to slow down a bit the ramp to the next product to not disrupt that demand. At this point of time, the (inaudible) is getting a customer. We made a choice, as been explained before, to ship it initially at lower throughput, so that the qualification could start earlier at our customer, so that they can be ready basically for high volume manufacturing next year when I think this is the real expectation for this tool to run manufacturing.

So from my point of view, maybe we said that the biggest element to help you is that technology wise, there is no concern at all. And when it comes to qualification with our customer, I think this is underway. And of course, we are going to use the next few months to mature the system. It always takes a few months to do that.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Thank you. And for Roger, I think if I'm not mistaken, in the last earning conference call, you talked about a EUR700 million of fast shipment. Did I hear you correctly that you said it's up to EUR1 billion now? Is that the right figure for fast shipment?

A - Roger Dassen {BIO 15064806 <GO>}

Yeah. I think there are different numbers, Mehdi, and that's why I just lumped it together. So, as I mentioned, we have a fast shipment effect, and your memory serves you well. But in addition to that, we also have 3800 that we shipped in H1, but recognizing revenue in H2. And also, we have High NA tools that with the long lead times and given this is a very new technology, we're going to recognize revenue not upon shipment, but upon SAC[ph]. And it's in that combination of things that we expect around 1 billion to be added to the -- to be added to the revenue. So deferred revenue that we expect to see in -- to see recorded in revenue in the second half. So that's the difference. It's not just a fast shipment, but also the other dynamics that I just talked about.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

You're welcome.

Operator

Thank you. Your next question comes from the line of Sandeep Deshpande from JP Morgan. Please go ahead.

Q - Sandeep Deshpande {BIO 3869012 <GO>}

Yeah, hi. Thank you for letting me on. My first question, I mean, clearly there was earlier questions on this, but I'm trying to ask it another way, which is that you don't have a view on where you're going to land in 25 years. And one can understand that given deep EUV, et cetera, they have much shorter lead times. But on EUV itself, given where the lead times on EUV are, you probably have a view on how many tools you're going to build into EUV already. So, I mean, my question would be, what is the capacity you have for EUV next year? And I mean, are you building fully to that capacity, or are you going to be building less than that capacity at this point? And I have a quick follow-up.

A - Roger Dassen {BIO 15064806 <GO>}

An A for effort, Sandeep, for sure in asking the question. But we're going to stick to our guns here, also on EUV. Wait until the Capital Markets Day. There we will give you a lot more color and context. We have been building capacity for EUV. As you know, we are working towards a capacity level of 90 in the '25, '26 time frame. So we have been driving up capacity quite a bit. So there is quite some room that we have to accommodate a demand on EUV. But we are not going to give any more color as to what our expectations for EUV sales are going to be. In November of 2025, we'll give you a more concrete picture. Just reiterating, between 30 billion and 40 billion, and it will not be the low point. I think that's the expectation that we've articulated before, and that's where we still are.

Q - Sandeep Deshpande {BIO 3869012 <GO>}

Thank you. Maybe stepping back a little bit. I mean, there have been a lot of conversations in the industry about that whether litho is going to continue the kind of scaling it has done over the last 20 years, given the challenges with shrinking here from the current level. So maybe in terms of the adoption of High NA and litho itself, Christophe, maybe you can make any comments on what your interactions with the foundries and the memory companies have been in terms of using litho as a critical driver of the scaling in semiconductors over the next few years because there has been a lot more conversation in the industry about the limitations of litho.

A - Christophe Fouquet {BIO 20524437 <GO>}

Well, I think it's two things that are well known. So, first, I think you're hinting to it. I think if we look at the last few years, the scaling, the shrink is slowing down. I think this is not new. And for a long time, litho had an impact on that. I think at this point of time, it's more fundamentals, basically, and related to all the innovation you need to put in place in order to be able to scale the transistor.

Now, this being said, scaling is slowing down. So that's many years already. What is not slowing down is the need for transistor density. So that curve, which is more slow, practically, is not slowing down. So customers are asking more and more density at the same speed they have done for many, many years. And as scaling slowed down, I would

say shrink become even more valuable because shrink is still today the best way to save cost on density. Everything else require you to just build more transistor with more steps. So shrink today, and that's true today, it's going to be true to more is still the best way to reduce cost. And sometimes, we say the less you can shrink, the more the value of shrink is going to be. That's one thing.

The other thing, of course, is if you shrink less, you need more volume. And more volume can be obtained by exposing more wafer or exposing more stack on the wafer. But in both cases, this drive also more lithography. So we never really feared the reduction of shrink because this was driving volume. This is one of the reasons why we have been building capacity. And at the same time, we know that every step we will be able to make in lithography will have huge value for our customers. So this is something also we believe continues to be important.

And of course, most probably over time and we'll also talk a bit about that in November, we will look at more ways to help our customer to drive this density basically up. So I say, no news there. Scaling is slowing down. It's not a difficult thing to say for a litho person, it's just reality. Density does not therefore need for more volume, more value for shrink. And I will say some opportunity to look at other applications to help our customers to guess this density at cost. We'll talk a lot about that in November, I like to say, because I think it's one of the things we want to explain again, everyone.

Q - Sandeep Deshpande {BIO 3869012 <GO>}

Thank you, Christophe.

A - Skip Miller {BIO 20244900 <GO>}

All right. We have time for one last question. If you were unable to get through on this call and still have questions, please feel free to contact the ASML Investor Relations department with your question. Now, operator, may we have the last caller, please?

Operator

Thank you. The last question today comes from the line of Adithya Metuku from HSBC. Please go ahead.

Q - Adithya Metuku {BIO 17642884 <GO>}

Yeah. Good morning. Good afternoon, guys. Thank you for taking my questions. So, firstly, just a clarification on your backlog. I just wondered if you could give us a sense for how big domestic Chinese demand is as a part of your backlog today, just to get a sense for the level to which it may contribute to revenues next year.

And secondly, just on China again, there's a lot of rumors flying around, and I suspect the rules that are in place today are the result of some give and take in different companies, different stakeholders pushing in different ways. But just to hear it from the horse's mouth, can you give us some idea of what the US government would ideally like you to ship to China regardless of the rules that are in place today? And how does that compare to what you're shipping to China today? Any color on that would be helpful. Thank you.

A - Roger Dassen {BIO 15064806 <GO>}

Adithya, on the backlog, I think we said it before that our backlog is a little bit north of 20%, and that's still the case. So that's on the Chinese share in our backlog.

On the China rumors, we're not going to add, I think, to the rumors that are already out there. I think this is a negotiation between governments. Companies are not at the table. And we live by -- we obviously stick to what we're allowed to do, and we're not going to speculate about what government A might want to see in the rules and what government B we might want to have in the rules. I don't think that's going to add to any clarity. I think what is very important to understand, and Christophe said it at the beginning of the call, the way we look at the demand for our tools is not from a specific geography. In this case, China.

We look at -- and that's the way we model our sales medium term and long term. We look at what is the global demand for wafers, and whether those wafers are being produced in country X or country Y, at the end of the day, it doesn't matter. And I think that is very important to recognize in looking at your model. We don't have a China element in our models. It is the global demand for wafers that drives our modeling.

Q - Adithya Metuku {[BIO 17642884 <GO>](#)}

Understood. Thank you, Roger.

A - Skip Miller {[BIO 20244900 <GO>](#)}

All right. Now, on behalf of ASML, I would like to thank you all for joining us today. Operator, if you could formally conclude the call, I'd appreciate it. Thank you.

Operator

Thank you. This concludes the ASML 2024 second quarter financial results conference call. Thank you for participating. You may now disconnect.

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